Project Report - SP-12 Riser R-3 Integrity Assessment

Field	Value
Document ID	PROJ-01
Revision	1
Date	2026-06-30
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Owner node	sp12_inspection_team
Classification	INTERNAL

Project summary

Item	Value
Project name	SP-12 Riser R-3 Integrity Assessment
Owner node	sp12_inspection_team
Domain	Asset integrity / external corrosion
Completion status	Phase 1 complete, phase 2 in progress
Related executors	Dr. Leila Moradi, Reza Kamali
Related knowledge items	INS-01, INS-02 (rev 2), scanned 2021 baseline

Research goals

- Measure the corrosion growth rate on riser R-3 in the splash zone.
- Estimate the remaining life against the 17.5 mm minimum wall.
- Compare repair options and recommend one.

Documents used

2021 baseline survey (scanned paper record), 2024 survey INS-01, 2026 survey INS-02 revisions 1 and 2, the riser design basis, and the Region South RBI plan.

Experiments and analyses

Readings from 2021, 2024 and 2026 were aligned by elevation and clock position. A linear growth model gave 0.63 mm per year at EL +8.5 m and about 0.9 mm per year at EL +6.5 m. Coating adhesion pull-off tests on two coupons gave 1.1 and 1.4 MPa, well under the 5 MPa specified for the original system.

Implementation activities

- Rope access survey added to the 2026 campaign.
- Temporary splash zone wrap specified and ordered.

Achievements

- First corrosion growth rate for R-3 based on three surveys.
- Remaining life estimate accepted by the regional integrity group.

Fictional test fixture. Region South Gas Operator is not a real company.

Challenges

- The 2021 baseline exists only as a scanned paper record.
- Readings in 2021 used different clock position labels, so two points could not be matched.
- Weather windows for rope access are short in winter.

Unresolved questions

- Is the growth rate linear, or is it speeding up as the coating fails?
- Can a clamp-on sleeve be installed without a production shutdown?